David Calvert, Ph.D.

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Summary: AI/ML Engineer with a Ph.D. in computational astrophysics and experience developing **LLM evaluation frameworks, predictive models, and scientific software**. Combines quantitative research with **Python, C++, PyTorch, and AI coding agents** to build, evaluate, and deploy AI systems.

Experience

AI/ML Engineer June 2026 - Present
Atomionics San Francisco, CA

- Developed **CNN models with CBAM attention** to predict ore locations from 2D raster data, achieving **0.92 ROC-AUC** and **2–14× lift in mineral discovery success** among the top 1% of ranked predictions on held out datasets.
- Built **evaluation frameworks and performance metrics** for predictive ML models and an LLM-based "AI Geologist" that retrieves documents from a vector database to recommend drilling locations alongside ML predictions
- Developed **backend infrastructure on Azure** using Docker containers and GitHub Actions workflows
- Implemented and developed a front-end UI used by the business development team alongside clients

Computational Astrophysicist 2019-2025
UC Berkeley / NC State University Berkeley, CA | Raleigh, NC

- Modeled three-dimensional supernova dynamics using a **Fortran multiphysics simulation code** on distributed high-performance computing systems
- Analyzed **terabyte-scale simulation outputs in Python**, extracting features and identifying structure in high-dimensional, nonlinear systems
- Adapted numerical solvers for nonlinear partial differential equations to preserve magnetic-field constraints on a three-dimensional Cartesian grid
- Developed **C++ linear algebra routines** to simulate quantum neutrino flavor oscillations in a one-dimensional supernova model

Software Engineer 2024 - 2026
KOINSLOT Inc. - GitHub San Francisco, CA

- Developed complete games, custom sprite-rotation algorithms, and asset-generation scripts in **C++** for *Kywy*, a handheld educational coding device

Independent Software Developer 2025
Kondra Chemical San Francisco, CA

- Designed, built, tested, and delivered a **LabVIEW application** to measure chemical concentrations in water samples
- Implemented hardware control, spectrometer data acquisition, and concentration interpolation modules

Business Systems Analyst 2018 - 2019
Insightsoftware Raleigh, NC

- Planned and executed CRM data migrations for multiple acquired companies, using **SQL and SOQL** to migrate records into Salesforce

Education

Ph.D., Computational Astrophysics Dec. 2025
UC Berkeley / North Carolina State University Berkeley, CA & Raleigh, NC

B.S., Physics with Honors & Applied Mathematics May 2018
North Carolina State University Raleigh, NC

Skills

Languages & Libraries: Python, C++, Fortran, SQL, MATLAB; PyTorch, NumPy, SciPy, scikit-learn, mpi4py
AI-Assisted Development: Claude, Codex, Omnigent, Muse; multi-agent workflows with Claude Teams
ML & Evaluation: CNNs, attention mechanisms, LLM evaluation frameworks, model benchmarking, performance metrics, statistical modeling
Infrastructure & Tools: Azure, Docker, GitHub Actions, Git, Linux, MPI, OpenMP, CUDA